

Undefined Thesis Paper Project

A Thesis
Presented to
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of the Requirements for the Degree
Bachelor of Arts

Riley Wood

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Approved for the Division
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Greg Anderson

Angélica Osorno

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Introduction

Chapter 1

The First

1.0.1 References and Labels

Sometimes you'd like to refer to a table or figure, e.g. you can see in Figure ?? that you can rotate figures . Start by labeling your figure or table with the label command (`\label{labelvariable}`) below the caption (see the chapter on graphics and tables for examples). Then when you would like to refer to the table or figure, use the ref command (`\ref{labelvariable}`). Make sure your label variables are unique; you can't have two elements named "default." Also, since the reference command only puts the figure or table number, you will have to put "Table" or "Figure" as appropriate, as seen in the following examples:

As I showed in Table ?? many factors can be assumed to follow from inheritance. Also see the Figure ?? for an illustration.

1.0.2 Footnotes and Endnotes

You might want to footnote something.¹ Be sure to leave no spaces between the word immediately preceding the footnote command and the command itself. The footnote will be in a smaller font and placed appropriately. Endnotes work in much the same way. More information can be found about both on the CUS site.

1.1 Bibliographies

Of course you will need to cite things, and you will probably accumulate an armful of sources. This is why BibLaTeX was created. For more information about BibLaTeX and bibliographies, see our CUS site (web.reed.edu/cis/help/latex/index.html)². There are three pages on this topic: *biblatex* (which talks about using BibLaTeX, at [/latex/biblatex.html](http://latex/biblatex.html)), TODO STYLE DOCUMENTATION and *bibman* (which covers how to make and maintain a bibliography by hand, without BibLaTeX, at [/latex/bibman.html](http://latex/bibman.html)). The last page will not be useful unless you have only a

¹footnote text

²reedweb:2007

few sources. There used to be APA stuff here, but we don't need it since I've fixed this with my apa-good natbib style file.

1.1.1 Tips for Bibliographies

1. Like with thesis formatting, the sooner you start compiling your bibliography for something as large as thesis, the better. Typing in source after source is mind-numbing enough; do you really want to do it for hours on end in late April? Think of it as procrastination.
2. The cite key (a citation's label) needs to be unique from the other entries.
3. When you have more than one author or editor, you need to separate each author's name by the word "and" e.g.
`Author = {Noble, Sam and Youngberg, Jessica},.`
4. Bibliographies made using BibTeX (whether manually or using a manager) accept LaTeX markup, so you can italicize and add symbols as necessary.
5. To force capitalization in an article title or where all lowercase is generally used, bracket the capital letter in curly braces.
6. You can add a Reed Thesis citation³ option. The best way to do this is to use the phdthesis type of citation, and use the optional "type" field to enter "Reed thesis" or "Undergraduate thesis". Here's a test of Chicago, showing the second cite in a row⁴ being different. Also the second time not in a row⁵ should be different. Of course in other styles they'll all look the same.

³noble:2002

⁴noble:2002

⁵reedweb:2007

Chapter 2

Tables and Graphics

2.1 Tables

The following section contains examples of tables, most of which have been commented out for brevity. (They will show up in the .tex document in red, but not at all in the .pdf). For more help in constructing a table (or anything else in this document), please see the LaTeX pages on the CUS site.

2.2 Figures

2.2.1 Common Modifications

The following figure features the more popular changes thesis students want to their figures. This information is also on the web at web.reed.edu/cis/help/latex/graphics.html.

Figure 2.1: Subdivision of arc segments. You can see that $p_3 = p'_6$.

Conclusion

Here's a conclusion, demonstrating the use of all that manual incrementing and table of contents adding that has to happen if you use the starred form of the chapter command. The deal is, the chapter command in $\text{\LaTeX}$ does a lot of things: it increments the chapter counter, it resets the section counter to zero, it puts the name of the chapter into the table of contents and the running headers, and probably some other stuff.

So, if you remove all that stuff because you don't like it to say "Chapter 4: Conclusion", then you have to manually add all the things $\text{\LaTeX}$ would normally do for you. Maybe someday we'll write a new chapter macro that doesn't add "Chapter X" to the beginning of every chapter title.

4.1 More info

And here's some other random info: the first paragraph after a chapter title or section head *shouldn't be* indented, because indents are to tell the reader that you're starting a new paragraph. Since that's obvious after a chapter or section title, proper typesetting doesn't add an indent there.

Bibliography

- [1] H. Verge, “A note on chernikova’s algorithm”, Apr. 2002.
- [2] P. Cousot and R. Cousot, “Comparing the galois connection and widening/narrowing approaches to abstract interpretation”, in *Programming Language Implementation and Logic Programming*, M. Bruynooghe and M. Wirsing, Eds., Berlin, Heidelberg: Springer Berlin Heidelberg, 1992, pp. 269–295, ISBN: 978-3-540-47297-1.
- [3] O. Kupferman, “Automata theory and model checking”, in *Handbook of Model Checking*, E. M. Clarke, T. A. Henzinger, H. Veith, and R. Bloem, Eds. Cham: Springer International Publishing, 2018, pp. 107–151, ISBN: 978-3-319-10575-8. DOI: 10.1007/978-3-319-10575-8_4 [Online]. Available: https://doi.org/10.1007/978-3-319-10575-8_4
- [4] G. J. Holzmann, “Explicit-state model checking”, in *Handbook of Model Checking*, E. M. Clarke, T. A. Henzinger, H. Veith, and R. Bloem, Eds. Cham: Springer International Publishing, 2018, pp. 153–171, ISBN: 978-3-319-10575-8. DOI: 10.1007/978-3-319-10575-8_5 [Online]. Available: https://doi.org/10.1007/978-3-319-10575-8_5
- [5] A. Chakarov and S. Sankaranarayanan, “Probabilistic program analysis with martingales”, in *Computer Aided Verification*, N. Sharygina and H. Veith, Eds., Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 511–526, ISBN: 978-3-642-39799-8.
- [6] E. Jones, bibinitperiod Main, C. Li, J. Marriott, and bibinitperiod Kavvos, “Domain-theoretic semantics for functional logic programming”, English, *Proceedings of the ACM on Programming Languages*, vol. 10, no. POPL, pp. 1641–1672, Jan. 2026, Publisher Copyright: © 2026 Owner/Author.; 53rd ACM SIGPLAN Symposium on Principles of Programming Languages (POPL 2026), POPL 2026 ; Conference date: 11-01-2026 Through 17-01-2026, ISSN: 2475-1421. DOI: 10.1145/3776699 [Online]. Available: <https://popl26.sigplan.org/>
- [7] D. O. Kidney and N. Wu, “Hyperfunctions: Communicating continuations”, *Proceedings of the ACM on Programming Languages*, vol. 10, pp. 174–203, 2026. [Online]. Available: <https://api.semanticscholar.org/CorpusID:283885146>

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- [8] É. Mangel, P.-A. Mellies, and G. Munch-Maccagnoni, “Classical notions of computation and the hassegawa-thielecke theorem”, *Proc. ACM Program. Lang.*, vol. 10, no. POPL, Jan. 2026. DOI: 10.1145/3776715 [Online]. Available: <https://doi.org/10.1145/3776715>
 - [9] N. Najmaei, N. van der Weide, B. Ahrens, and P. R. North, “From semantics to syntax: A type theory for comprehension categories”, *Proc. ACM Program. Lang.*, vol. 10, no. POPL, Jan. 2026. DOI: 10.1145/3776725 [Online]. Available: <https://doi.org/10.1145/3776725>
 - [10] P. B. Levy and M. Rogers, “What is a monoid?”, *Proc. ACM Program. Lang.*, vol. 10, no. POPL, Jan. 2026. DOI: 10.1145/3776727 [Online]. Available: <https://doi.org/10.1145/3776727>